AI ARCHITECTURE — DETAILED DESIGN

Predictive, Allocation & Feedback Intelligence

ATLANTECH · INTELLIGENT PORT OPERATIONS

AI Architecture v3 · September 2026

All AI learning stays on the port's own server

RUNTIME

Staff action (dashboard) → Laravel (modular monolith) → Redis queues (queue:part1 · queue:part2) → Python workers (one per part) → MySQL (result written) → SSE (dashboard updates)

Python scheduler (periodic pulls (A1) · no Laravel call) · Part 3 (local · reads / writes MySQL)

Berthing Request
Preliminary Approval recorded

Customs Approval
new status · entered manually by staff

on-demand pull

A1 · Data Ingestion

- Weather · wave · current — every 6 h
- On-demand weather + surge at Customs Approval
- Astronomical tide — daily, multi-day horizon
- Surge input — follows the weather schedule
- External marine data via adapter

→ **environment_forecast (MySQL)**

ETL

PART 1 — PREDICTIVE & ENVIRONMENTAL

produces constraints & entry-time proposal · never allocates resources

A2 · Astronomical tide
Harmonic model from published constants (pyTMD) · in-house

PHYSICS

A3 · Surge residual
Surge(t) = SeaLevel(t) − Tide(t)
a definition, not a prediction

FORMULA

A4 · Threshold Feasibility
wind · gust · visibility · wave · tide/surge vs operating limits
exact & auditable safety rules

RULES

A5 · UKC
Under-keel clearance + Barrass squat
naval-architecture

PHYSICS

A7 · Demand / Peak forecast (M3)
Phase 1: aggregate open Berthing Requests + history
Phase 2: LightGBM regressor

ML · PHASE 2

A6 · Uncertainty margin
statistical margin or provider ensemble spread

STATISTICS

Trigger Customs Approval → queue:part1 → Part 1 worker

Hard safety limits (wind · wave · visibility · UKC) are fixed and never loosened automatically

Part 1 output

```
feasible_windows start · end · confidence
blocked_periods start · end · limiting_factor
peak_congestion_score
data_versions weather · tide
```

persisted to MySQL · pushed to dashboard via SSE

HUMAN GATE 1
Port Authority Staff reviews & confirms the entry-time window
→ ENTRY_WINDOW_CONFIRMED · becomes Part 2 input

confirmed window + constraints

corrected inputs → Part 1

BAPLIE upload (staff)

B1 · BAPLIE Parser
Deterministic EDIFACT parser
UN/EDIFACT D.17A · SMDG 2019
→ vessel · containers · stowage
all-or-nothing commit

DETERMINISTIC

PART 2 — AI ENGINE

sole authority for shared-resource allocation · berth · tug · pilot

B2 · Constraint filter
LOA · draft · terminal type · safety-officer constraints

RULES

B3 · Service time (M2)
Phase 1: formula
Phase 2: LightGBM quantile P50 / P90

ML · PHASE 2

B4 · Delay risk (M4)
LightGBM classifier on deviation & override history

ML

B5 · CP-SAT solver
OR-Tools · Berth Allocation Problem · guaranteed feasible

OPTIMIZATION

B6 · Rationale
Fixed templates tied to binding constraints · never an LLM

TEMPLATE

Inputs confirmed window + Part 1 constraints
BAPLIE-derived vessel & cargo data
live (actual) berth · tug · pilot state
Marine Safety Officer constraints

B7 · Re-planning
re-solve · rolling horizon

OPTIMIZATION

Trigger
window confirmed → queue:part2 → worker → MySQL → SSE

Part 2 output — ranked options

```
rank · berth_id · start · end
tugs · pilot_id
service_time p50 · p90
delay_risk · rationale_text
```

HUMAN GATE 2
Staff confirms the Operational Plan or overrides a component
→ reason stored in recommendation_override

new model version

delay / cancel → re-plan

PART 3 — FEEDBACK LOOP

local to this port only · no model, weight or update ever leaves this server

C4 · Forecast bias correction
regression / quantile mapping on the forecast-vs-actual gap

STATISTICS

C3 · Retraining pipeline
time-based split · Champion vs Challenger · guarded automatic

MLOPS

C2 · Drift monitor
PSI + rolling MAE on M2 · M3 · M4

STATISTICS

C1 · Outcome evaluation
MAE · override rate · wait time · Part 1 window accuracy

STATISTICS

forecast vs actual

GUARDED AUTOMATIC RETRAINING

Retrains periodically with no approval step → the Challenger replaces the Champion only if it is not worse on MAE and override rate (held-out, time-based data)
→ otherwise the Champion keeps running and a flag is logged for review · overrides are analyzed as a signal, never a silent retraining trigger

Operational Plan · PLAN_CONFIRMED

Execution
existing systems & personnel

Actual outcomes

```
actual_status_update (manual)
deviation_log
recommendation_override
```

LEGEND — platform / system of record · human decision · AI part (Python satellite) · machine-learning model (M2 · M3 · M4) · non-ML: physics · rules · optimization · statistics

Out of scope this round: Vision module · cross-port / SaaS model sharing (rejected)